

cF1bÒ .c Jcv) bvDd Æ

dv0br 'dFEc6*df/ d!4c - b†'dF*d 5d!D

1. bvDdVCd†F bvDb63bv3d¢ r 6ö

dVDd Ö .tsx dvH bvDdVCd†F bvDdV3bFHdB 9db 1c6l

```
'use client';
import { MapContainer, TileLayer, Marker, Popup } from 'react-leaflet';
import 'leaflet/dist/leaflet.css';
import L from 'leaflet';
```

b6HdB 3cv1 'use client' cf1d†1d¢ Db6F Leaflet d!-b!bÂ %dFI Browser

b†Fc6*d†1bò B Ed6Hdfb

- MapContainer: bvDbÖ'd†Jb' 'dF#c6'c6Jb' DdF.
- TileLayer: d!-dVQdB (dF'cv'b¢ 'dF.c Jc
- Marker: c-DbvEb' †/b
- Popup: df'd 0b' Edf(b

1.1 bV5dF'bÒ #d!l

dv0dr EcFCdF) dV9c Hd) dV9 Webpack d, æPxt.js. Leaflet

```
delete (L.Icon.Default.prototype as any)._getIconUrl;
L.Icon.Default.mergeOptions({
  iconUrl: 'https://unpkg.com/leaflet@1.9.4/dist/images/marker-icon.png',
  iconRetinaUrl: 'https://unpkg.com/leaflet@1.9.4/dist/images/marker-icon-2x.png',
  shadowUrl: 'https://unpkg.com/leaflet@1.9.4/dist/images/marker-shadow.png',
});
```

dfEc6- bvDbæ'cVJb' 'd

1.2 b†JbvFb

```
const craftsmen = [
  { id: 1, name: 'b6-dV/ - d6Gc (bv&d¢rÂ ÷6-F-öã¢ ³3 ã CCBÂ 3 ã#3Suò 2 ¶çVÖ&W"Â ç',
    { id: 2, name: 'dV-dV/ - c6(bvC', position: [30.0500, 31.2400] as [number, number],
    { id: 3, name: 'c-Dd¢ Ò FbÆ'c rÂ ÷6-F-öã¢ ³3 ã 3f Â 3 ã#3 Ò 2 ¶çVÖ&W"Â çVÖ&W%Ö
  ];
```

dv0dr EcVAd†Ab' *bÖ*d†J c-Dd' 2 AdfJd!F

dVDbv-c†): as [number, number] cf1d†1d¢ Dd TypeScript dF#db &V 7BÖ

1.3 dvJd6D bvDbæ1

```
<MapContainer
  center={[30.0444, 31.2357]}
  zoom={13}
  style={{ height: '100%', width: '100%' }}
>
  <TileLayer
    attribution='&copy; <a href="https://www.openstreetmap.org/">OpenStreetMap</a>'
    url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png"
  />
  {craftsmen.map((c) => (
    <Marker key={c.id} position={c.position}>
      <Popup>{c.name}</Popup>
    </Marker>
  ))}
</MapContainer>
```

- center: dfBcv) bvDdV

- zoom: 13 dV3b!Hd' *d&1
- style: bvDbæ1d!7b' *b6.c R E
- TileLayer: d!3b!.böE OpenStreetMap dV,bv
- Craftsmen.map: dfEc 9dFI d6D d Fd¢ Hdf6c' Ddr Ö ke

2. bvDbv3b!/c-!b Ad¢ ge.tsx — bvDd

- d J cVAbÖ) bvDd Æ æF-ær ‡ vP.tsx) `Â 'dF.c Jcv) b!Oc
- ```
const Map = dynamic(() => import('@components/Map'), {
 ssr: false,
 loading: () => (
 <div className="bg-[#dbeee0] h-80 md:h-96 rounded-2xl
flex items-center justify-center text-gray-500 text-sm">
 bÆ'c J b!-dVJ
 </div>
),
});
```
- dynamic(): dVJc&) dVF Next.js b!3dVI "Dynamic Import" — d6Hbò 'dF.c Jcv) dF' d!O
  - ssr: false: dv0br Gd, 'dF#dvE. d!Edf9 b!-dVJdB 'dF.c Jcv) c-Dd' 'dF3d!1d 1
  - loading: bö'dF) b!Oc ,c' ¥5, Jc†Gc #b¶Fbv! b!-dVJdB 'dFEd6Hc

---

## 3. d†6c' 'dF.c Jcv)

- d J JSX cVAbÖ) bvDd Æ æF-æ
- ```
<section className="bg-gradient-to-l from-[#0a2e1f] to-[#0f5132]">
<div className="max-w-6xl mx-auto px-6 py-16 grid md:grid-cols-2 gap-10 items-center">

<div className="relative rounded-2xl overflow-hidden order-2 md:order-1 h-80 md:h-96">
  <Map />
  <div className="absolute bottom-3 right-3 left-3 bg-white
rounded-xl shadow-sm p-3 flex items-center gap-3 z-[1000]">
    <div className="w-9 h-9 rounded-full bg-[#eef6ef] ...">Ø=Þáp </div>
    <div>
      <p className="text-sm font-bold text-gray-900">dVHb¶B 100%</p>
      <p className="text-xs text-gray-500">dV-b!1d Hdb Ed
    </div>
    </div>
  </div>
  <div className="order-1 md:order-2 text-right">...</div>
</div>
</section>
```
- bvDbæ1d!7b' Ad
- order-2 md:order-1: d J bvDdVHb†'d!D bvD
 - rounded-2xl overflow-hidden: c&HbvJbr 'dF.c Jcv) dVBcV
 - Badge bvDb6EbvF: cF1d!7 b6(d!6 cFAbvA d J b63d D bvDbæ1c
 - z-[1000]: dF6dV'db 8dvHc 'd

4. b!/d B b

- c-FböEbr Jd *bÖ 'dFEc6*
- Next.js d!(bö# b†*bÖEd!D bvDcVAb
 - d!8dv1 dVCd†F bvDb!-dVJd
 - b†9bò 'd6*dV'dB ‡-G ation d J bvDc
 - MapContainer d!8dv1 d J bvDd DôÖ Hd!(bö# b†,d
 - 3 dV'c Cc 'b¢ *c†
 - c-Fbò 'dF6c!7 c-Dd' #d¢ Ö

5. bvDbv9b!Ebv/d!'b¢ „Fv

dVF package.json:

- leaflet: dVCb!(b' 'd
- react-leaflet: React wrapper dF@ Leaflet.
- @types/leaflet: TypeScript definitions.

6. dFEbv0

- b†/d†F API Key: OpenStreetMap dV,bvFd¢ (bvDd6'dV
- b!-dVJdB Cd !: Dynamic Import d!Edf9 b!-
- c6Gd†Db' 'dF*d†3c“ dF'c6*b†/l
- dV*d†'d B dV9 RTL: b